

DAYAPURAM RESIDENTIAL SCHOOL

SHORT NOTE – DATA HANDLING

BAR GRAPH:

- A display of information using bars of uniform width, their heights being proportional to the respective values.

DOUBLE BAR GRAPH:

- A bar graph showing two sets of data simultaneously. It is useful for the comparison of the data.

CIRCLE GRAPH(PIE-CHART)

- A circle graph shows the relationship between a **whole and its parts**.
- The whole circle is divided into **sectors**.
- The size of each sector is **proportional** to the activity or **information** it represents.
- The formula to find the centre angle of each activity:

$$\text{center angle} = \frac{x}{\text{total}} \times 360$$

- The formula to find the portion of activity if the angle is given,

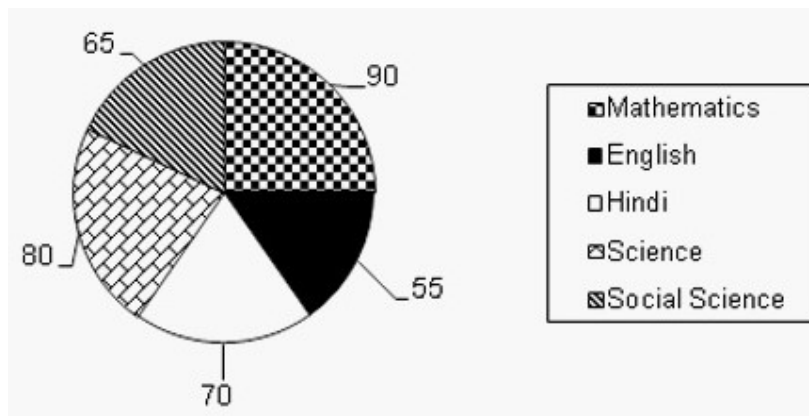
$$\text{portion of particular activity}(x) = \frac{\text{center angle}}{360} \times \text{total}$$

Questions:

1. The graph used to compare two sets of data is:
a) Bar graph b) double bar graph c) circle graph d) pictograph
2. In a circle graph, the parts are represented as:
a) Chords b) segments c) sectors d) Arcs
3. A group of people is asked about their favourite subject. The data obtained is shown in the following table. Draw a pie chart.

Subject	Maths	Science	Malayalam	English
Number people	12	45	20	23

4. A survey is conducted among 180 people to check the most favourite subject.



- a) The total central is _____
- b) To draw the pie chart, the whole circle is divided into _____
- c) Which subject is liked by most of the students
- d) Which is the least favourite subject?
- e) The number of people liked Hindi is _____

PROBABILITY:

$$\text{probability} = \frac{\text{number of favourable outcome}}{\text{total number of outcomes}}$$

Questions:

1. A die is thrown once.
 - a) What is the probability of getting an odd number?
 - b) What is the probability of getting a prime number?
 - c) What is the probability of getting a composite number?
2. What is the probability that a number selected from the numbers 1,2,3,4,...,20 is not a multiple of 3?
3. In a basket, there are **4 blue** balls, **5 red** balls, and **1 green** ball. One is chosen from the balls without looking.
 - a) What is the probability of getting a red ball?
 - b) What is the probability of getting a non-green ball?
 - c) What is the probability of getting a ball that is neither red nor green?
4. In a school playground, some balls are kept in a basket for a game. There are **10 footballs, 6 basketballs, and 8 volleyballs**. A student is asked to pick **one ball at random** from the basket.
 - a) Find the probability that the student picks a **basketball**.
(i) 3/12 (ii) 10/24 (iii) 8/24 (iv) 4/24
 - b) Find the probability that the ball picked is **not a volleyball**.
(i) 5/12 (ii) 7/12 (iii) 10/24 (iv) 16/24
 - c) Find the probability that the student picks **either a football or a volleyball**.
(i) 18/24 (ii) $\frac{3}{4}$ (iii) $\frac{2}{3}$ (iv) 10/24
 - d) Find the probability that the student picks **neither a football nor a basketball**.
(i) 8/24 (ii) $\frac{1}{3}$ (iii) $\frac{5}{12}$ (iv) $\frac{1}{2}$